

Lecture 18: Delay time tomography

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Outline and Motivation

We now extend ray theory to image 3-D variations in the Earth's structure.

- We will introduce **delay time tomography**, a technique to create quantitative images of lateral velocity variations inside the Earth.

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- We will introduce **delay time tomography**, a technique to create quantitative images of lateral velocity variations inside the Earth.
- Such tomographic models provide our primary observational constraint on processes like mantle convection.
- This is a classic **linear inverse problem**, and we will discuss a practical method for solving it known as **regularised least squares**.

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Delay Time

We define the delay time, δT , as the difference between the observed travel time (T) and the time predicted by a reference spherical Earth model (T_0).

$$\delta T = T - T_0$$

A positive δT means the wave arrived late, indicating it travelled through a region that is, on average, slower than the reference model. These delay times are our data.

Relating Delay Times to Velocity (1/2)

The relationship between travel time T and velocity v is non-linear. However, since the observed delay times are small, we can assume the Earth's true velocity (v) is only a small perturbation (δv) away from the reference velocity (v_0).

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Fermat's Principle

Recall Fermat's Principle: the travel time of a ray between two points is stationary with respect to small perturbations of the ray path.

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The Linearised Relationship

Therefore, to first order, the delay time is simply the integral of the slowness perturbation ($\delta(1/v) \approx -\delta v/v_0^2$) along the **unperturbed ray path** in the reference model.

$$\delta T = - \int_{\text{ray}_0} \frac{\delta v}{v_0^2} ds$$

Setting Up the Inverse Problem (1/2)

Our goal is to find the continuous velocity perturbation function, $\delta\alpha(\mathbf{x})$, from a finite set of delay time measurements, $\{\delta T_i\}$.

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- This is a fundamentally **underdetermined** problem.
- To make it solvable, we **parameterise** the model. We assume the velocity perturbation can be represented as a weighted sum of a finite number of known basis functions, $\varphi_j(\mathbf{x})$.

Model Parameterisation

$$\delta\alpha(\mathbf{x}) = \sum_{j=1}^m \delta\alpha_j \varphi_j(\mathbf{x})$$

The unknowns are now the m coefficients, $\delta\alpha_j$.

Setting Up the Inverse Problem (2/2)

Substituting the parameterised model into our linearised relationship for each measurement i gives:

$$\delta T_i = \sum_{j=1}^m \left(- \int_{\text{ray}_i} \frac{\varphi_j}{\alpha_0^2} ds \right) \delta \alpha_j + e_i$$

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This is a system of linear equations.

Matrix Formulation

We can write this compactly in matrix-vector form:

$$\mathbf{d} = \mathbf{A}\mathbf{m} + \mathbf{e}$$

- $\mathbf{d}$ is the $n \times 1$ data vector of delay times, δT_i .
- $\mathbf{m}$ is the $m \times 1$ model vector of unknown coefficients, $\delta \alpha_j$.
- $\mathbf{e}$ is the vector of measurement errors.
- $\mathbf{A}$ is the $n \times m$ sensitivity matrix (or kernel), where

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- If $\mathbf{m}$ is a solution, then $\mathbf{m} + a\mathbf{m}_0$ is also a solution for any scalar a , because $\mathbf{A}(\mathbf{m} + a\mathbf{m}_0) = \mathbf{A}\mathbf{m} + a\mathbf{A}\mathbf{m}_0 = \mathbf{d} + \mathbf{0} = \mathbf{d}$.

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Physical Meaning of the Null Space

The null space corresponds to structures in the Earth that our set of rays does not sample. For example, a small velocity anomaly in a region where no rays pass cannot be detected and is part of the null space.

Solving the Problem (1/2): Regularisation

To overcome non-uniqueness, we need to add more information or constraints.

- A standard least-squares approach minimises the data misfit:

$$J_{data}(\mathbf{m}) = (\mathbf{A}\mathbf{m} - \mathbf{d})^T \mathbf{C}^{-1} (\mathbf{A}\mathbf{m} - \mathbf{d})$$

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- We use **regularisation** by adding a second term that penalises models we consider "undesirable" (e.g., models that are too large or too rough).

Regularised Misfit Function

We minimise a combined objective function:

$$J(\mathbf{m}) = \underbrace{(\mathbf{A}\mathbf{m} - \mathbf{d})^T \mathbf{C}^{-1} (\mathbf{A}\mathbf{m} - \mathbf{d})}_{\text{Data Misfit}} + \underbrace{\lambda \mathbf{m}^T \mathbf{B} \mathbf{m}}_{\text{Model Norm / Penalty}}$$

Here, λ is a trade-off parameter.

Solving the Problem (2/2): The Solution

To find the model $\mathbf{m}$ that minimises the regularised misfit function $J(\mathbf{m})$, we set its gradient with respect to $\mathbf{m}$ to zero.

$$\nabla J(\mathbf{m}) = \mathbf{0}$$

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This yields a system of linear equations for $\mathbf{m}$.

The Normal Equations

$$(\mathbf{A}^T \mathbf{C}^{-1} \mathbf{A} + \lambda \mathbf{B}) \mathbf{m} = \mathbf{A}^T \mathbf{C}^{-1} \mathbf{d}$$

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The Regularised Least-Squares Solution

Solving for $\mathbf{m}$ gives the unique solution:

$$\mathbf{m} = (\mathbf{A}^T \mathbf{C}^{-1} \mathbf{A} + \lambda \mathbf{B})^{-1} \mathbf{A}^T \mathbf{C}^{-1} \mathbf{d}$$

The regularisation term $\lambda \mathbf{B}$ ensures the matrix is invertible.

A Bayesian Perspective

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- $p(\mathbf{m})$ is the **prior**: what we believe about the model before seeing the data.
- $p(\mathbf{d}|\mathbf{m})$ is the **likelihood**: the probability of observing our data given a specific model.
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Connection to Least Squares

If we assume the prior knowledge and the data errors are both Gaussian, then the model that maximises the posterior probability is **identical** to the regularised least-squares solution.

- The regularisation term $\lambda\mathbf{B}$ is equivalent to assuming a Gaussian prior on the model parameters.

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- This system is non-unique due to a **null space**. It is solved using **regularised least squares**, which balances data fit with a model penalty.
- The regularised least-squares solution is equivalent to the most probable model in a **Bayesian** framework with Gaussian assumptions.